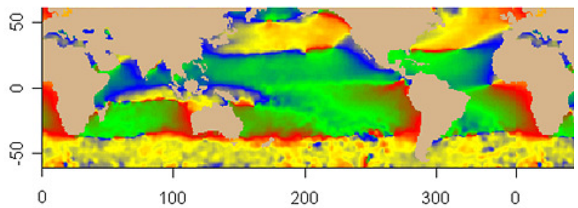
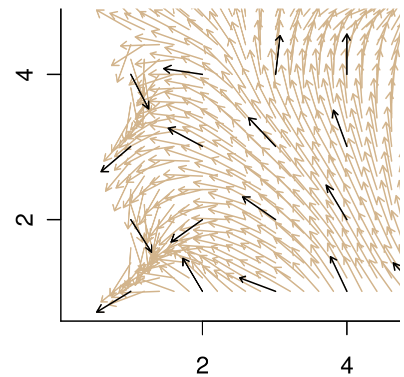
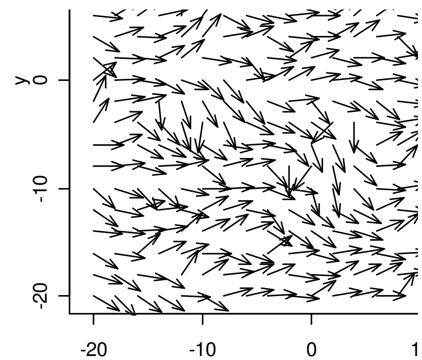
R Package CircSpatial for the

Imaging - Kriging - Simulation







# of Circular-Spatial Data

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Circular Random Variable (CRV)

Definition

- Takes random direction in a plane
- The total probability mass of all possible directions distributed on the unit circle.

PDF

- Typically plotted on a unit circle

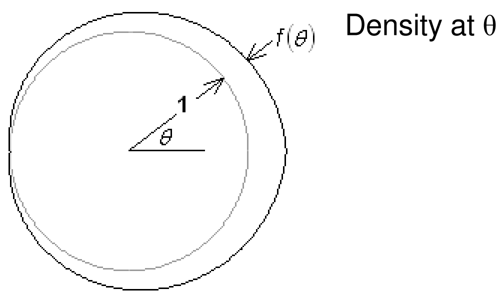
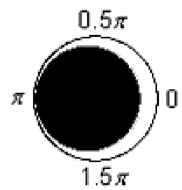
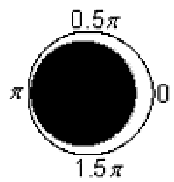
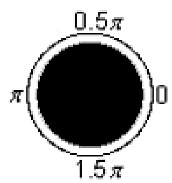
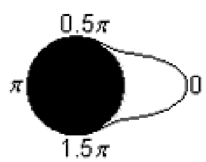
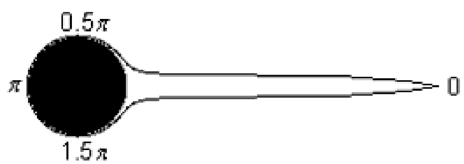


Figure 1. PDF of a Circular Probability Distribution Plotted on Outside of Unit Circle to Show Density vs. Angle.

Circular PDFs for $-\pi \le \theta \le \pi$

Name of Distribution	Circular PDF Plot	Circular PDF Function	Range of Parameter	Value of ρ in PDF Plot
Cardioid		$\frac{1}{2\pi}\left[1+2\rho\cos(\theta)\right]$	$0<\rho\le 0.5$	$\rho = 0.95 \times 0.5$
Triangular		$\frac{4-\pi^2\rho+2\pi\rho\lvert\pi-\theta\rvert}{8\pi}$	$0<\rho\le\frac{4}{\pi^2}$	$\rho = .95\times\frac{4}{\pi^2}$
Uniform		$\frac{1}{2\pi}$	NA	NA
von Mises		$\frac{\exp(\kappa\cos(\theta))}{2\pi\sum_{j=0}^{\infty}\left(\frac{\kappa}{2}\right)^{2j}\left(\frac{1}{j!}\right)^2}$	$0<\kappa<\infty$	$\kappa = 10.2696$ equivalent to $\rho = .95$
Wrapped Cauchy		$\frac{1}{2\pi}\frac{1-\rho^2}{1+\rho^2-2\rho\cos(\theta)}$	$0<\rho<1$	$\rho = 0.95 \times 1$

Analogous to the uniform RV

Analogous to the normal RV

Several circular pdf contain trig functions

Some Applications

Biology – Direction of migration

Geology - Fault orientation

Geophysics - Magnetic field direction

Meteorology - Wind direction

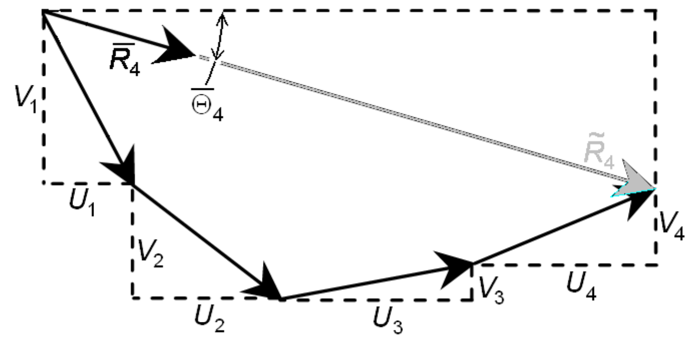
Oceanography - Ocean currents

Periodic Phenomena - Births/month, deaths/month, accidents/hour

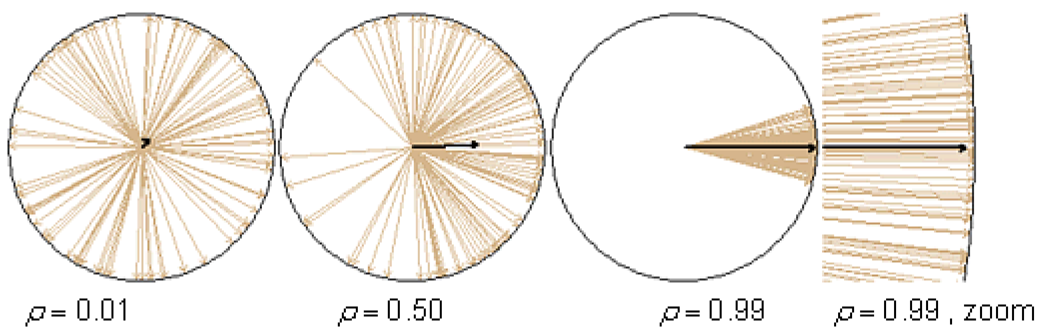
My Motivation – At ATK Space Launch Systems our data is measured on rocket motor parts with circular cross-sections

A Little Statistics

- **Mean**
  - Direction of the vector resultant of observations of direction as unit vectors



- **Concentration (New Term)**
  - Opposite the sense of variance
  - Length of black arrow computed as length of resultant / # Observations



Uniform –All directions equally likely, ρ =0

Degenerate – Only one directions likely, ρ =1

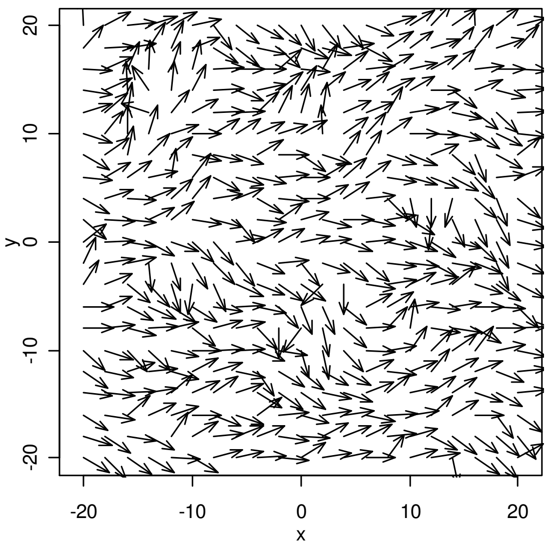
Random Field (RF)

RF Definition

- A stochastic process operating over a space containing RVs with spatial dependence such that variations from the mean direction tend to be more similar as distance between sample locations decreases.

Circular Random Field (CRF)

- Variable is random direction



Location of observation is (x,y) of tail of arrow

von Mises CRF with
Distribution Parameter ρ =.8
Spatial Parameter Range=10

R Package CircSpatial

How Is an R Package Made?

- "Writing R Extensions" in R Help/Manuals
  - Required Installations: Rtools, MiKTeX, HTMLWorkshop

For a Graphical Interface

- http://bioinf.wehi.edu.au/~wettenhall/RTclTkExamples/

Main Functions

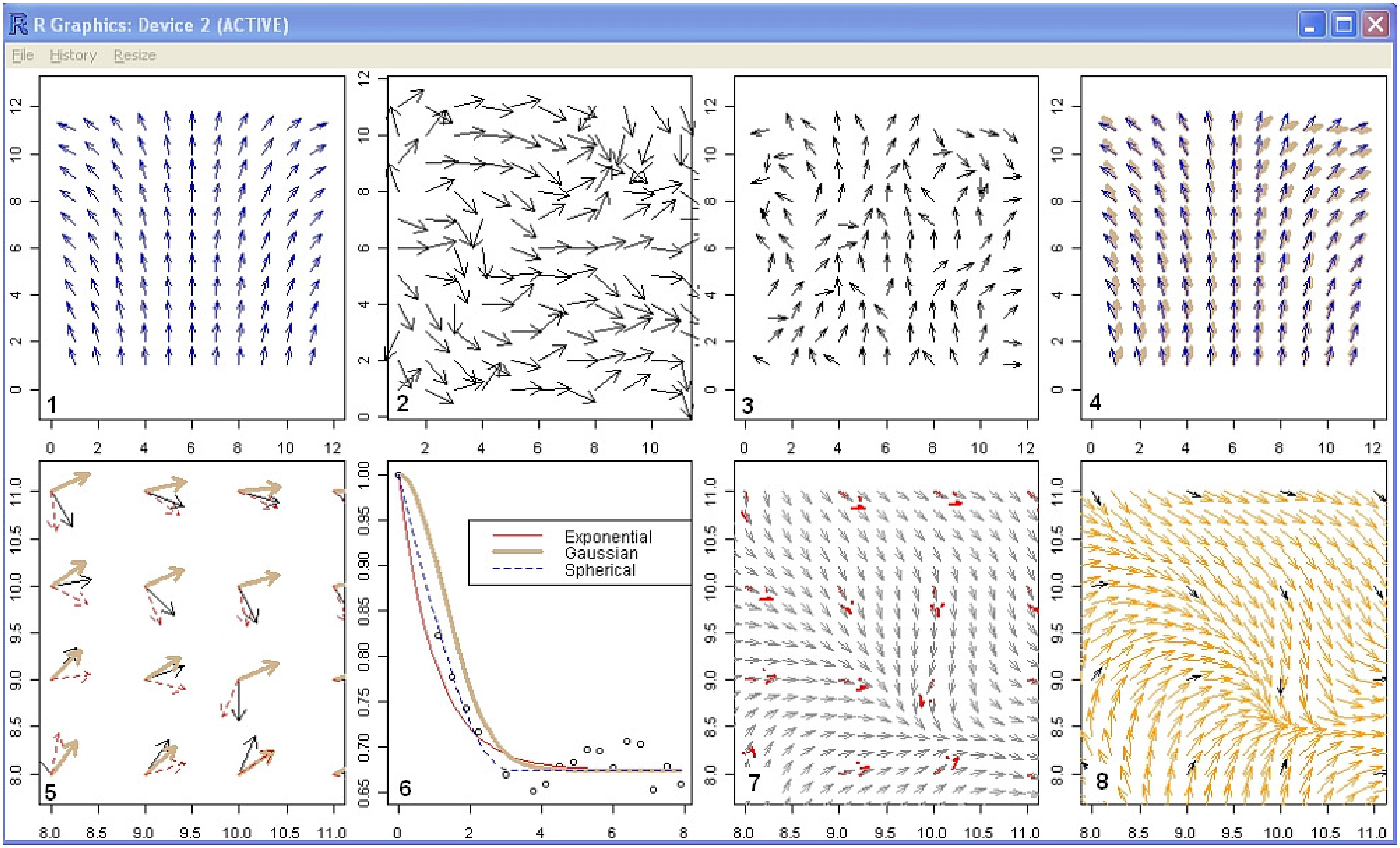
- **SimulateCRF**: Simulate a random field of CRV with spatial correlation
- **CircResidual**: Compute residual variations from the mean direction
- **CosinePlots**: Plot the empirical and fitted models of the spatial correlation
- **KrigCRF**: Estimate direction at an unsampled location using the spatial correlation model and residuals
- **InterpDirection**: Interpolate the estimated trend model of direction at an unsampled location
- **CircDataimage**: A GUI for interactive imaging of circular-spatial data
- **PlotVectors**: Plot vector-spatial data

Location of Software

- Not yet in CRAN

Comprehensive Example

The numbers in the lower left corner of the plots refers to the steps on the following pages



Comprehensive Example (1)

1 - Construct underlying trend using arrow.plot of R package fields

Note that direction rotates clockwise from West-to east trend

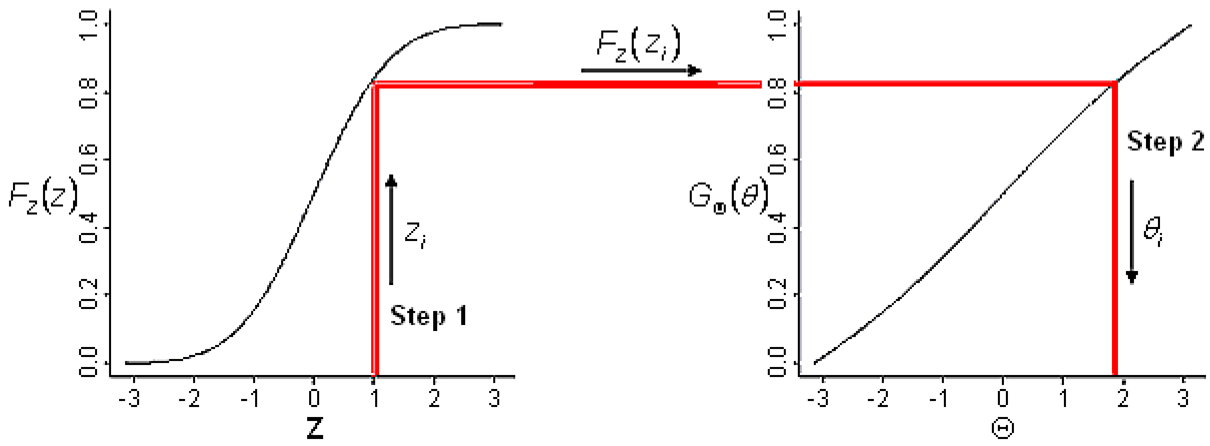
2 - Compute sample of a circular random field via function SimulateCRF

How does it work?

– Generate a sample of a GRF

$$\boxed{\begin{array}{l} \boldsymbol{Z} \sim N_n(\boldsymbol{0}, \Sigma = \boldsymbol{I}) \\ \mathrm{E}\{\tilde{\boldsymbol{C}}\boldsymbol{Z} + \boldsymbol{\mu}\} = \tilde{\boldsymbol{C}}\mathrm{E}\{\boldsymbol{Z}\} + \boldsymbol{\mu} = \tilde{\boldsymbol{C}}\boldsymbol{0} + \boldsymbol{\mu} = \boldsymbol{\mu} \\ \mathrm{Cov}(\tilde{\boldsymbol{C}}\boldsymbol{Z} + \boldsymbol{\mu}) = \tilde{\boldsymbol{C}}\,\mathrm{Cov}(\boldsymbol{Z})\tilde{\boldsymbol{C}}^T = \tilde{\boldsymbol{C}}\,\boldsymbol{I}\,\tilde{\boldsymbol{C}}^T = \tilde{\boldsymbol{C}}\tilde{\boldsymbol{C}}^T = \boldsymbol{C} \end{array}}$$

– Map an observation of a spatially correlated normal RV to circular RV via the normal CDF and circular inverse CDF



Comprehensive Example (2)

3 - Compute a sample with an underlying spatial trend

4 – Fit an appropriate model

Note fitted to cosines and sines separately to avoid the problem that

the 0 degree location and the 360 degree location are the same direction. What is the average direction? Not 180 degrees.

5 – Get the residuals via function CircResidual

What is the residual?

Analogous to a residual for a linear RV, Residual = Data – Spatial Trend

Why get residuals?

- The rotation of the data from the spatial trend codes the spatial correlation, i.e., are more similar as distance between sample locations decreases

Why get the spatial correlation?

- We will use spatial correlation to estimate direction at unmeasured locations

Design choice for closely related graphs

- Color will uniquely identify the same entities in a sequence of graphs

Comprehensive Example (3)

6 – Decode the spatial correlation

Function = CosinePlots

What does it do?

$$\hat{\varsigma}(d) = \frac{1}{N(d)} \sum_{\left| \| \boldsymbol{x}_j - \boldsymbol{x}_i \| - d \right| < \varepsilon} \cos(\theta_j - \theta_i)$$

Why does it work?

- The prediction which minimizes error depends on the mean cosine between observed directions as a function of distance between observations

What does the plot tell us?

- Describe the graph (axes, points, curves, range, sill)

- Manipulate the spatial parameters (range & sill) to get the curve of best fit to the points

- Best model is exponential, range=3.07, sill=0.674

$$\varsigma(d) = \begin{cases} 1, \ dist = 0 \\ sill + (1 - sill)\exp(-3\, dist/range), \ d > 0 \end{cases}$$

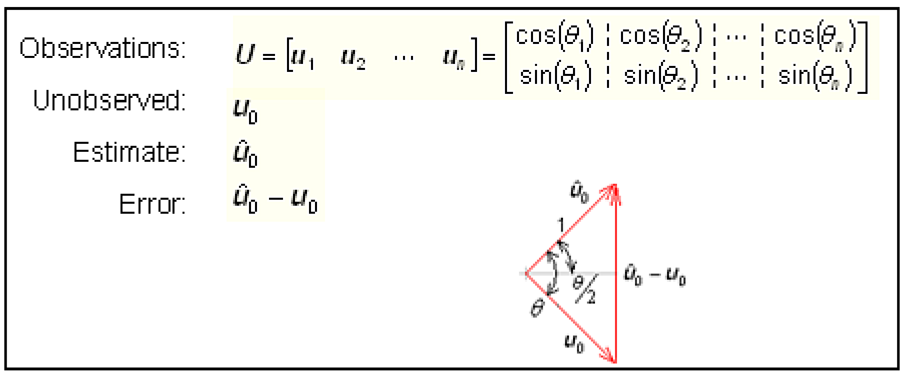
Comprehensive Example (4)

7 – Estimate direction using the spatial correlation model and function = KrigCRF

Why is it called Kriging?

How does it work?

- Finds the linear combination of observations that minimizes the error vector length.



- With 6 pages of trig and linear algebra, it’s

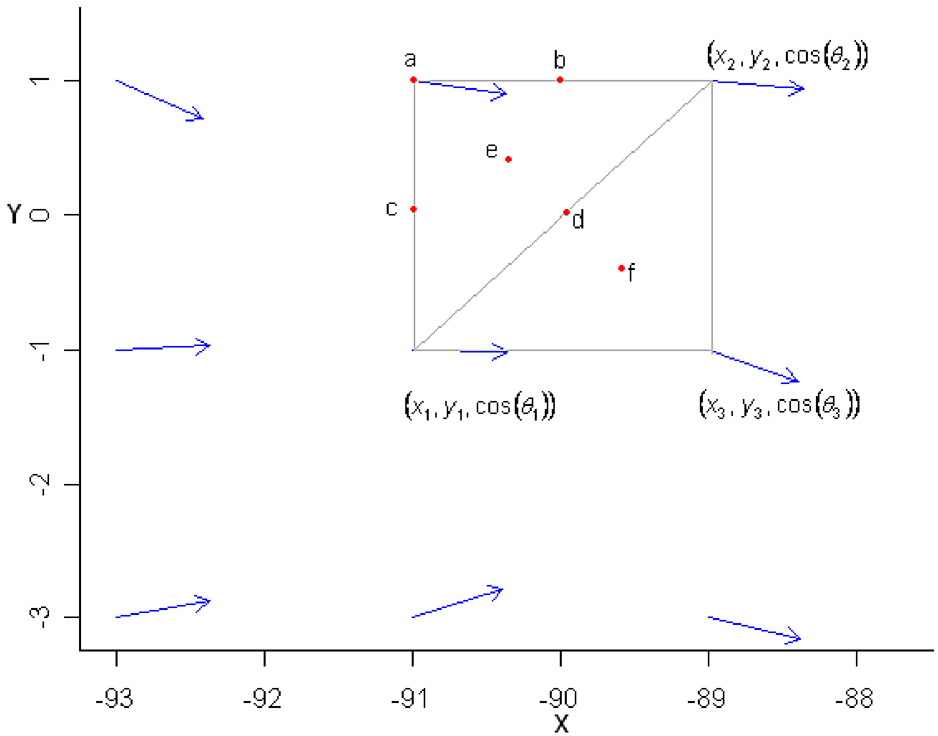
$$\boldsymbol{w} = \boldsymbol{K}^{-1}\boldsymbol{c} \Big/ \sqrt{\boldsymbol{c}^T \boldsymbol{K}^{-1} \boldsymbol{U}^T \boldsymbol{U} \boldsymbol{K}^{-1} \boldsymbol{c}}$$

Let’s do it

Comprehensive Example (5)

– To estimate the underlying spatial trend at unmeasured locations I use the function InterpDirection

How it works



Let's see it work

8 – Let's see what the estimated random and underlying spatial trend look like when combined?

Imaging Circular-Spatial Data (1)

Data

- Homogeneous ocean wind data subset from the International Comprehensive Ocean Atmosphere Data Set (ICOADS) at http://dss.ucar.edu/datasets/ds540.1/data/msga.form.html
- Covers 7 years x 4 months giving 0 to 28 observations per location
- 495,688 observations of month, year, longitude, latitude, and east and north components of wind velocity in 0.01 m/s

Function=PlotVectors

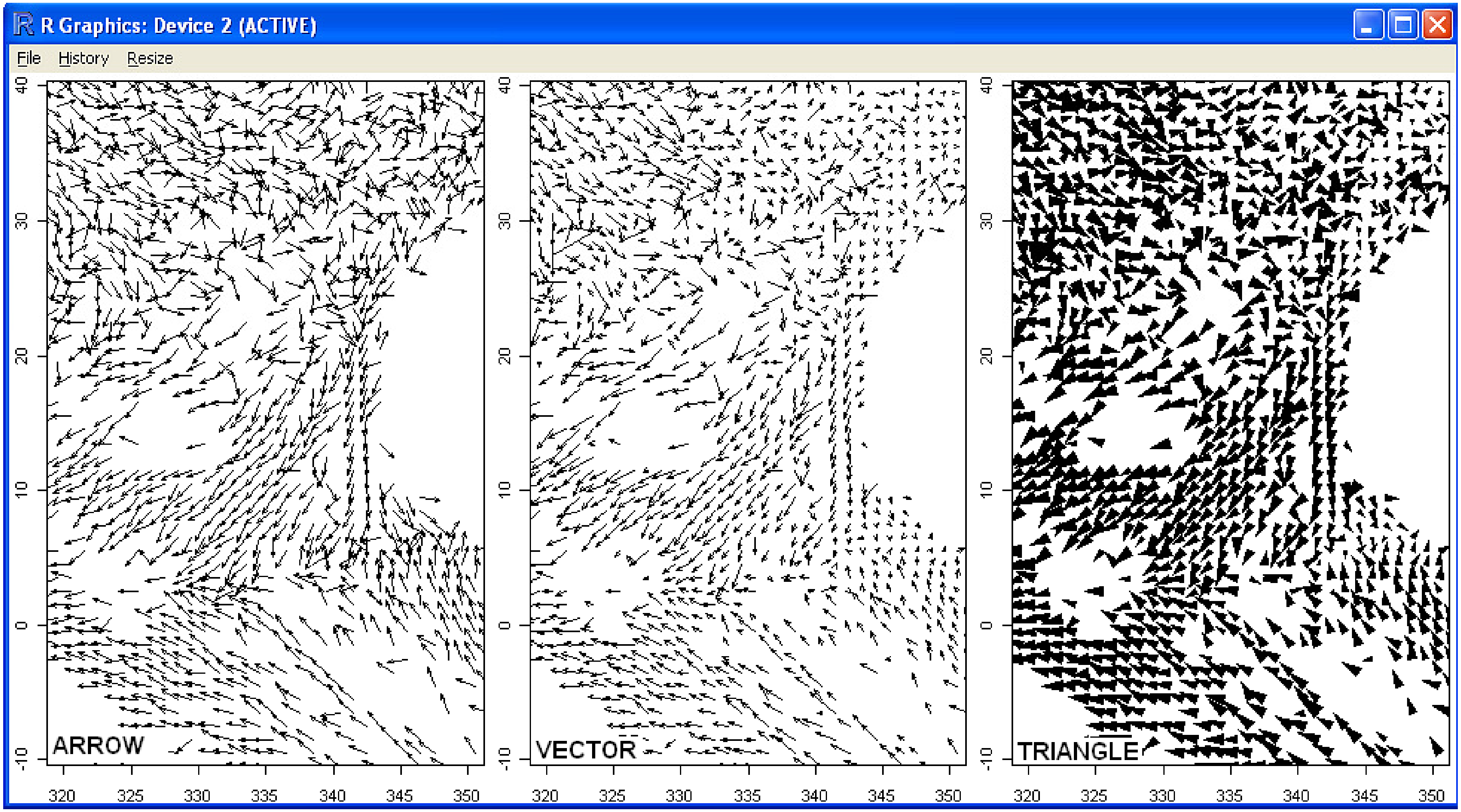
Types (Examples on next page)

- Arrow (Constant length)
- Vector (Length=magnitude)
- Triangle Icon (Area=magnitude)

Enhancement

- Jittering, addition of some noise to the location of the arrow helps with smooth data and models

Imaging Circular-Spatial Data (2)

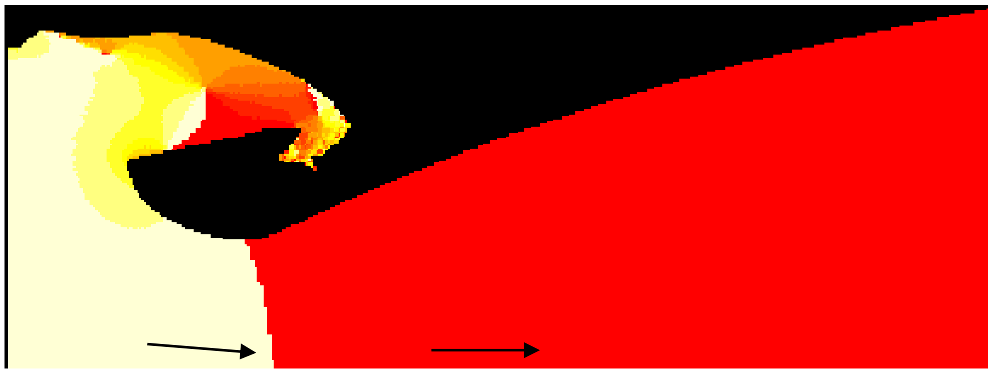


Color Wheel (1)

Problem – Arrow Plots become unintelligible at higher arrow density

Solution – Heatmap

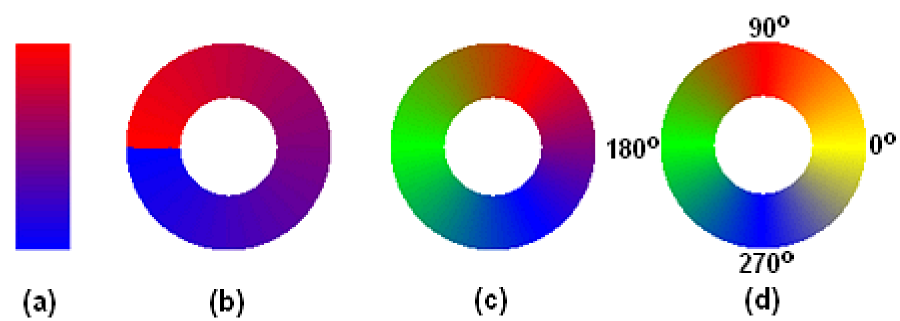
- Example, partial view of flow inside a rocket nozzle with heat colors for direction



Problem

- Image discontinuity occurs around cross over from 360 deg to 0 deg

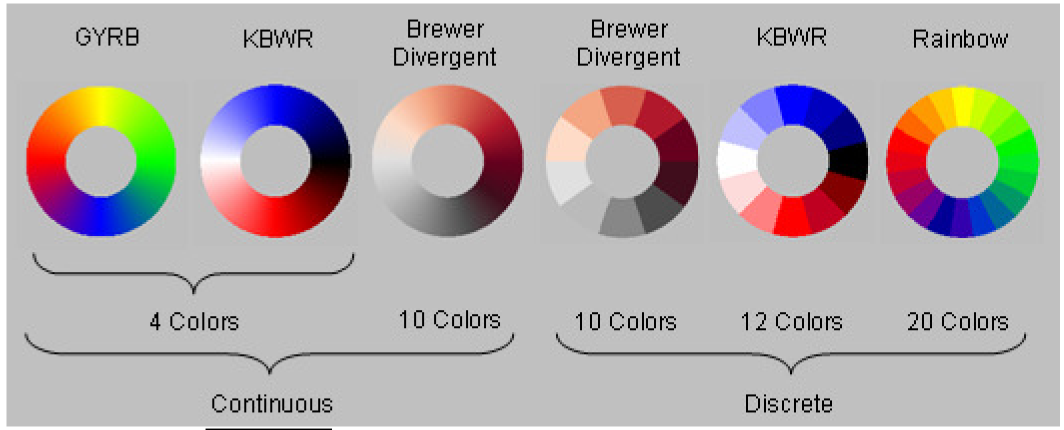
Solution – Color wheel



Evolution of color Wheel

Color Wheel (2)

Other Color Wheels



Detail

Areas of similar direction have same color

Function=CircDataimage

Motivation

- Initial version was menu driven and the menus were always presented in the same sequence
- To fully interact, the user must be able to change settings in any order

Demo

Interesting Features

- Structure west of Americas and Africa
- Vortex latitude [0,50], longitude [200,250]

Design

- Rotate color wheel to highlight interesting structure

Circular Dataimage

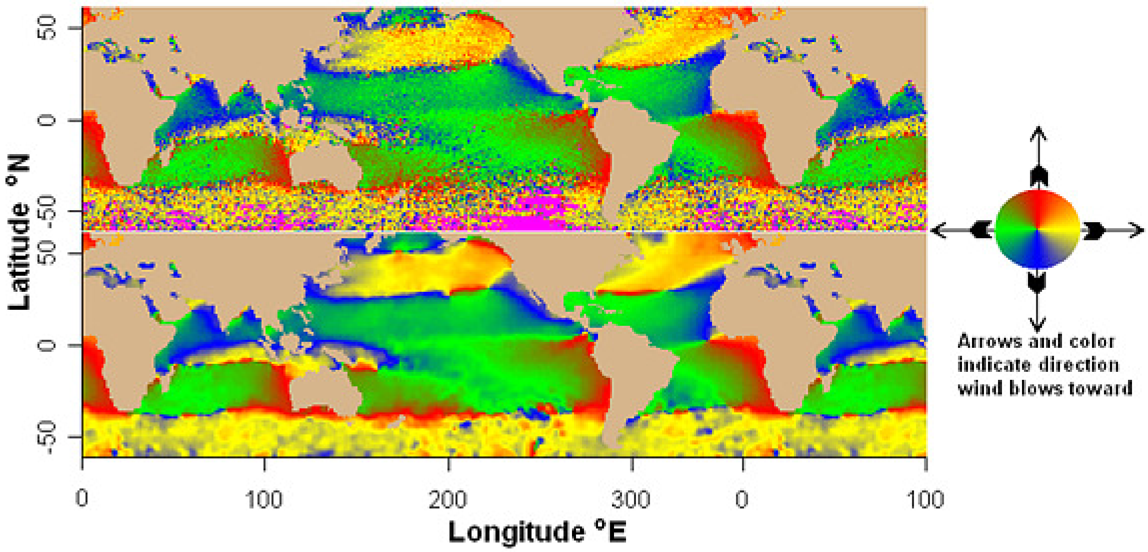


Figure 2-1. Circular Dataimages of the Direction Wind Is Blowing Toward, Coded with Yellow-Red-Green-Blue (YRGB) Color Wheel (Right). Average direction is shown in the top plot, and smoothed average direction in the bottom plot. The circular dataimage provides a continuous high resolution image of circular-spatial data with structure simultaneously recognizable on a broad range of scales.